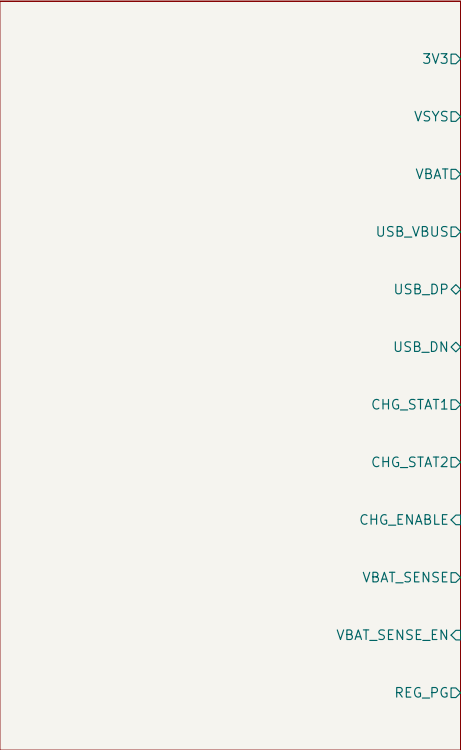


Cicala Rev B.05 system block diagram

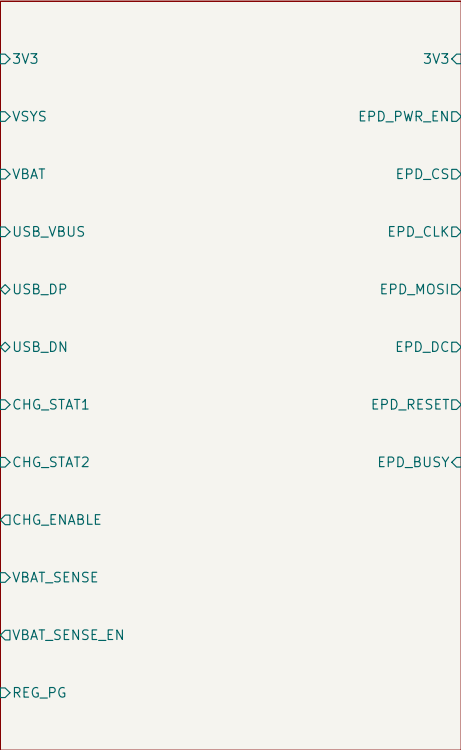
Power flow: USB—C 5 V or the protected cell -> BQ25185 power path (VSYSD)
-> TPS63802 buck-boost (3V3) -> TPS22917 load switches, one for the battery divider and one for the panel and its boost network.

Power and USB



File: sheets/battery_charger.kicad_sch

Controller and user I/O



File: sheets/esp32_core.kicad_sch

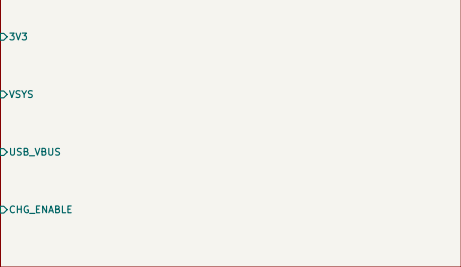
E-paper display



File: sheets/epaper.kicad_sch

GND is a global net and is not drawn here.

Charge temperature guard



File: sheets/charge_temperature.kicad_sch

Engineering prototype; physical qualification pending		
Cicala		
Sheet: /		
File: cicala_rev_b.kicad_sch		
Title: Cicala Rev B.05 schematic overview		
Size: A4	Date: 2026-09-16	Rev: B.05
KiCad E.D.A. 10.0.5	Id: 1/5	

USB-C provides 5 V and native USB, BQ25185 charges the protected cell; TP563802 regulates 3.3 V.

USB-C and protection

USB-C receptacle J1

VBUS A4 D-USB_VBUS

CC1 A5 B5

CC2 B5

D- A7 B7

D+ A6 B6

SH A1

USB_SHIELD

R33 0

C31 1nF 2kV DNP

SBUS1 B8

SBUS2 B8

5.1k R2

5.1k R1

10uF 10V C4

USB_VBUS

USBLC6-25C6 U4

I/O1 1

I/O2 3

I/O1 6

I/O2 4

GND 2

R3 22

R4 22

USB_DN

USB_DP

3.3 V buck-boost

0.47uH XFL4015-471ME L2

TP563802DLA U3

EN MODE 1

VOUT L2 6

VOUT 7

VOUT 3V3

VIN 9

AGND 10

FB 4

PG 5

D-REG_PG

VSYS-Q 1

GND 2

VSYS-Q 9

GND 10

511k 1% R14

91k 1% R15

100k R16

47uF 10V C8

22uF 6.3V C9

47uF 10V C10

100nF C11

3V3 D3V3

3V3 D3V3

3V3 D3V3

3V3 D3V3

3V3 D3V3

3V3 D3V3

Charger and protected cell

VBAT DVBAT

TP22 VBAT_CELL

R34 0

J3 VBAT_CELL

BATT_NTC

USB_VBUS-Q

BQ25185DLH U2

IN 10

TS/MR 6

ILIM/VSET 7

CE 4

ISET 8

GND 5

EP 11

SYS BAT 1

BAT 2

STAT2 3

STAT1 9

D-VSYS

D-VBAT

CHG_STAT2

CHG_STAT1

CHG_CE

CHG_ILIM_VSET

CHG_ENABLE

R7 18k

R13 4.7k

R8 3.00k

10uF 10V C6

10uF 25V C5

10k R10

10k R9

Battery + external 10k B3435 NTC

Switched battery measurement

VBAT

C34 10uF 10V

VBAT_SENSE_END

GND

TP522917DBVR U5

VIN 1

CT 4

ON 3

GND 2

QOD 5

VOUT 6

R11 1M

R12 470k

C7 100nF

DVBAT_SENSE

PWR_FLAG

PWR_FLAG

PWR_FLAG

VBAT-Q

USB_VBUS-Q

Engineering prototype; physical qualification pending

Cicala

Sheet: /Power and USB/

File: battery_charger.kicad_sch

Title: Power, charging, and USB

Size: A4 Date: 2026-09-16 Rev: B.05

KiCad E.D.A. 10.0.5 Id: 2/5

3.3 V buck-boost

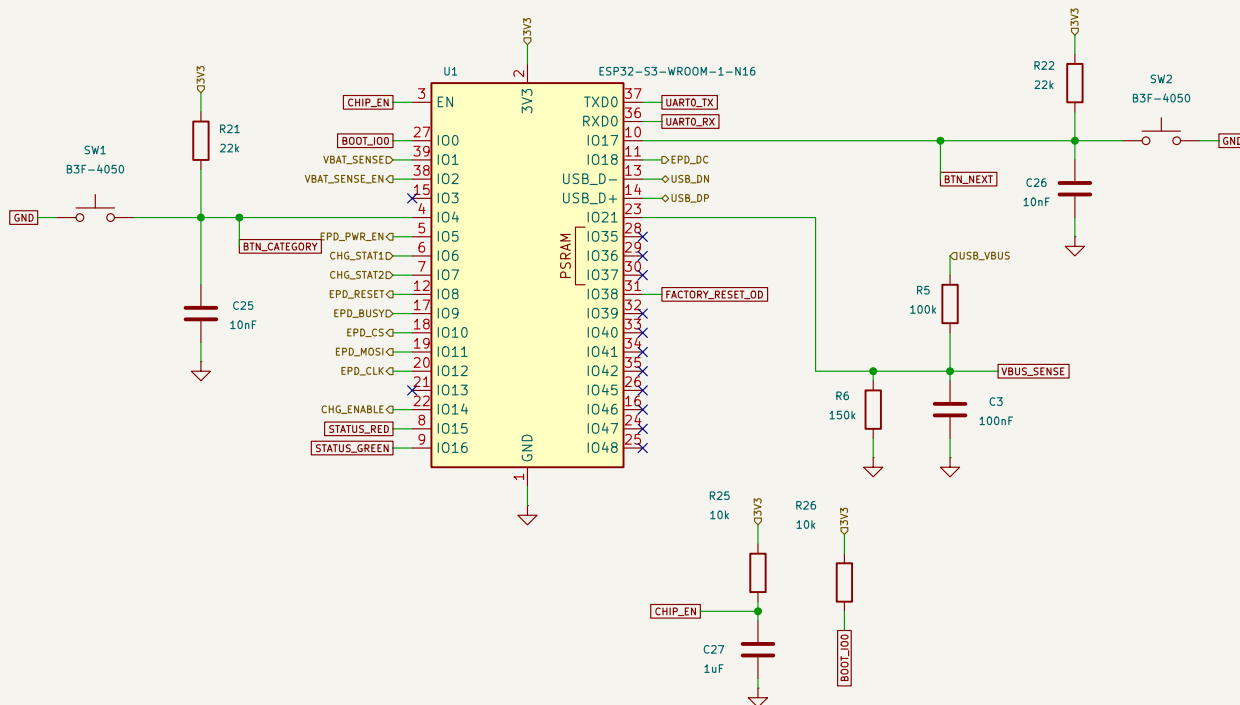
Switched battery measurement

The circuit diagram shows a TPS22917DBVR (U5) configured for switched battery measurement. The input VBAT is connected to the VIN pin (pin 1) and a 10uF capacitor (C34). The CT pin (pin 4) is connected to VBAT_SENSE_END. The QD pin (pin 5) is connected to GND. The VOUT pin (pin 6) is connected to a 1M resistor (R11) and a 470k resistor (R12) to ground. A 100nF capacitor (C7) is connected to the output node. The output is labeled DVBAT_SENSE.

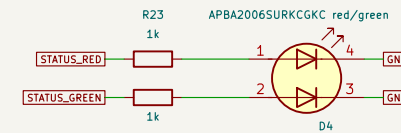
Sheet: /Power and USB/		
File: battery_charger.kicad_sch		
Title: Power, charging, and USB		
Size: A4	Date: 2026-09-16	Rev: B.05
KiCad E.D.A. 10.0.5		Id: 2/5

WROOM-1 is preferred; WROOM-1U remains the placement/RF fallback. USB is the normal debug path.

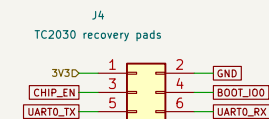
ESP32-S3 module



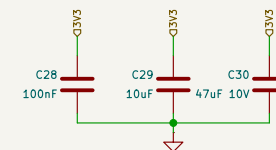
Status LED



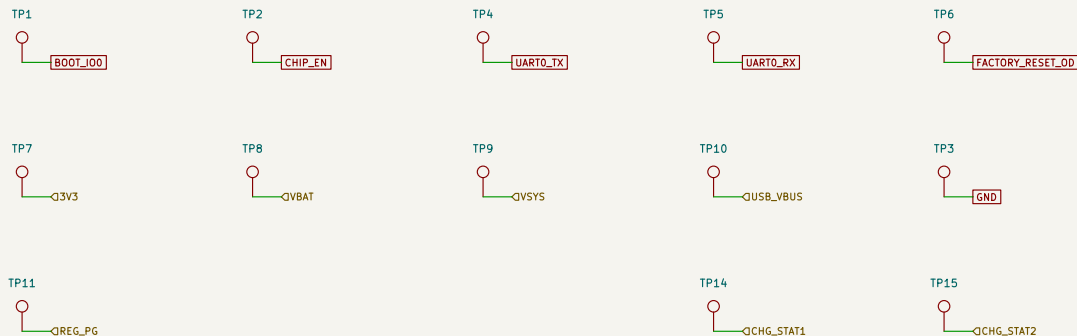
Recovery header



Module decoupling



Production test points



Engineering prototype; physical qualification pending
Cicala

Sheet: /Controller and user I/O/
File: esp32_core.kicad_sch

Title: Controller, user I/O, and recovery

Size: A4
KiCad E.D.A. 10.0.5

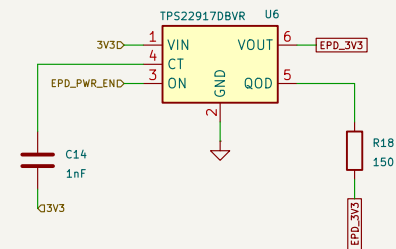
Date: 2026-09-16

Rev: B.05
Id: 3/5

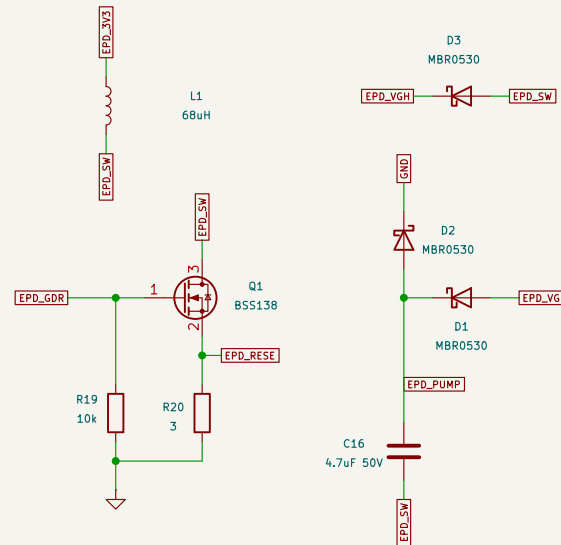
The load switch removes panel and boost power between refreshes.
Boost network follows Good Display's reference.

Panel pinout follows V1.1 table: 1/4 NC, 5 VDHR.
Confirm panel revision and rails at first bring-up.

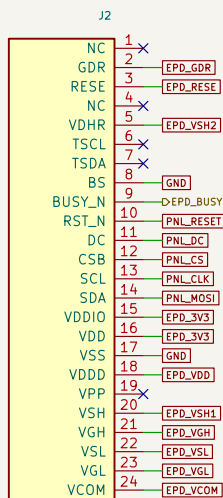
Switched panel supply



V1.1 pin table: pins 1/4 NC, pin 5 VDHR. HAT boost: 68uH / BSS138 / 3R.

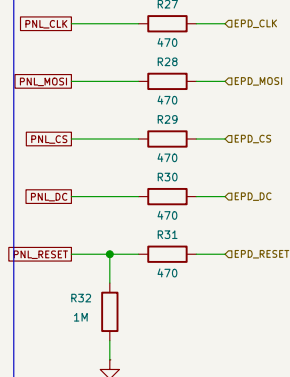


Panel FPC

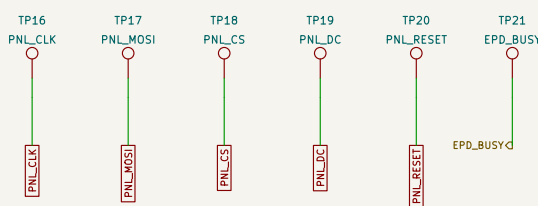


Waveshare 3.52in V1.1 UC8253

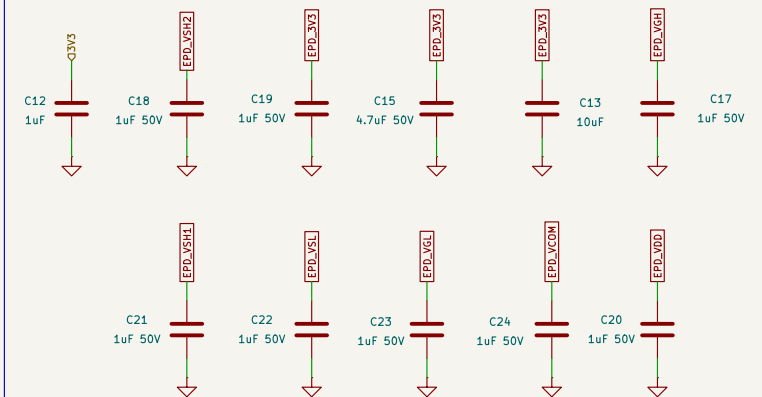
Bus isolation



Panel bus test points



Panel reservoir capacitors



Engineering prototype; physical qualification pending

Cicala

Sheet: /E-paper display/

File: epaper.kicad_sch

Title: UC8253 e-paper interface and boost

Size: A4

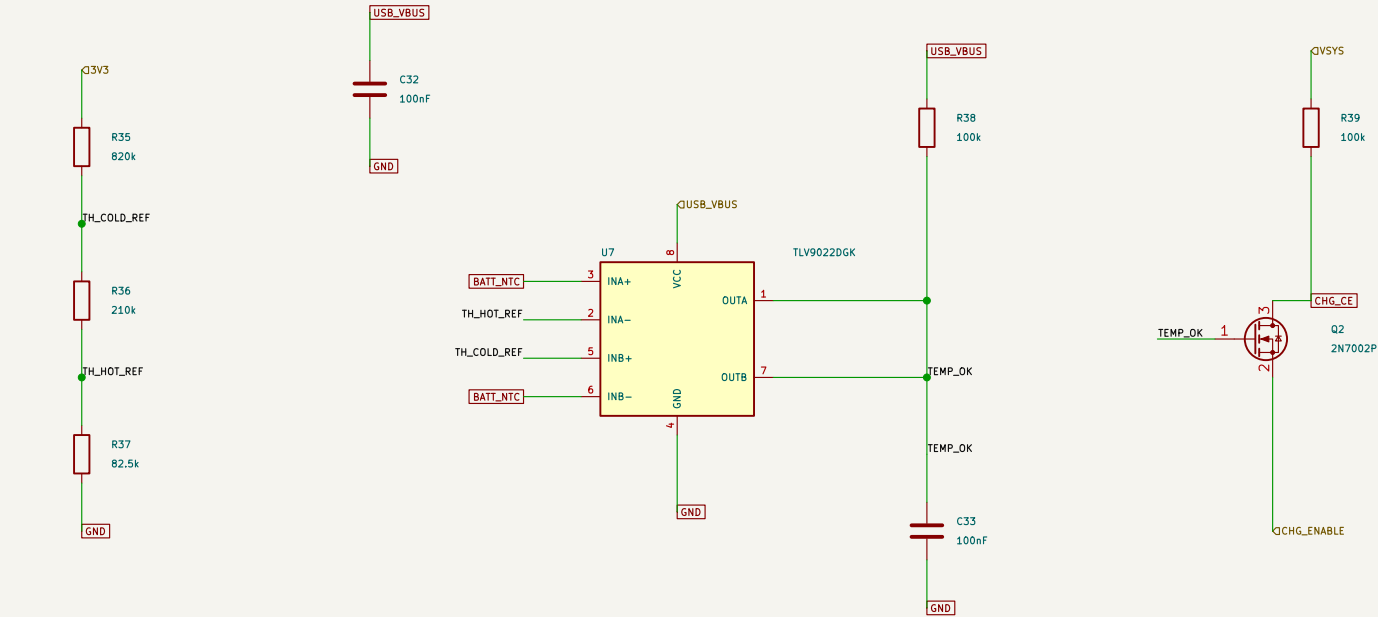
Date: 2026-09-16

Rev: B.05

KiCad E.D.A. 10.0.5

Id: 4/5

Independent battery temperature veto



J3.2 / BATT_NTC remains connected to BQ25185 TS/MR.
Semitec 103JT-025 is bonded to the protected cell and returns to J3.3.

Both comparator outputs must release TEMP_OK to permit charging.
An open or shorted thermistor disables charging. C33 delays permission at USB insertion.
Firmware CHG_ENABLE high disables charging; R13 supplies the low default through Q2.

3V3 sets the comparison thresholds; U7 and the gate pull-up run only from USB.
The 1.1125 Mohm reference ladder draws 2.97 uA from 3V3.

Engineering prototype; physical qualification pending
Cicala

Sheet: /Charge temperature guard/
File: charge_temperature.kicad_sch

Title: Hardware charge temperature guard

Size: A4	Date: 2026-09-16	Rev: B.05
KiCad E.D.A. 10.0.5		Id: 5/5